

SCADA, Meteorological and NWP Data Specification

LLM-readable plant data interface specification for multi-horizon forecasting

Document Type	Technical Data Specification
Version	v1.0
Use	Sample PDF upload for LLM requirement extraction, planning and forecast design
Data Status	Synthetic demonstration - not real plant SCADA or commercial data
Primary Objective	Enable a model to infer business use case, horizon, variables, methods, metrics and data gaps

LLM extraction note: This document intentionally uses clear labels, tables and repeated field names so a document-reading model can extract business objectives, SCADA/MET/NWP variables, forecast horizons, metrics and RFP requirements with minimal ambiguity.

1. Purpose and Scope

This document defines the SCADA, meteorological, NWP, grid and forecast output interfaces needed by a renewable forecasting solution. It is designed as a sample PDF upload for an LLM onboarding agent. The agent should read the fields, infer what is available, identify missing variables and generate a horizon-wise forecast plan.

LLM-readable context: DOCUMENT_TYPE = SCADA_MET_NWP_SPECIFICATION. PLANT = Pavagada Solar Park synthetic sample. HIERARCHY = grid -> plant -> hub -> block. USE = variable extraction + data gap analysis + forecast planning.

2. Plant Data Hierarchy

Level	Asset ID format	Description	Primary use
Grid	GRID_PAVAGADA_220KV	Pooling station / grid-node aggregate.	Grid schedule, reserve and dispatch decisions.
Plant	SOL_PAVAGADA	Whole solar park aggregate.	Plant-level forecast, deviation and business objective tracking.
Hub / Segment	PAV_HUB_1 ... PAV_HUB_8	Group of blocks with local sensors and event attribution.	Locate ramp/curtailment/data-quality issue.
Block	PAV_S3_B02 etc.	Granular inverter/sensor block.	SCADA, sensor and equipment diagnosis.

3. SCADA Tag Specification

Field name	Unit	Refresh	Required horizons	Definition / quality rule
actual_mw	MW	5min	5min,15min,2hour,40hour	Measured/exported power. Must refresh within expected latency; stale if unchanged for >2 intervals.
possible_power_mw	MW	5min	all horizons	Estimated power available if not constrained. Used to separate curtailment and weather effects.
scheduled_mw	MW	15min	15min,2hour,40hour	Committed dispatch-block schedule. Required for deviation penalty use case.
local_limit_mw	MW	5min	all horizons	Active grid/local export limit. Curtailment when possible power exceeds this and actual clips near limit.
control_setpoint_mw	MW	5min	5min,15min,2hour	Power plant controller target; explains operator/grid instructions.
availability_pct	%	5min	all horizons	Equipment availability. Low value indicates equipment derating rather than weather-only effect.
inverter_status	enum	5min	5min,15min	Inverter online/offline/derated status at block level.
scada_quality	enum	5min	all horizons	GOOD, SUSPECT, STALE. Used as quality gate before model update.
scada_latency_seconds	seconds	5min	5min,15min	Latency check; high latency reduces trust in real-time nowcast.
ramp_rate_mw_min	MW/min	5min	5min,15min	Recent power change rate used for ramp detection.
curtailment_status	enum	5min	all horizons	NONE, WATCH, CONFIRMED based on local_limit and possible/actual relationship.

4. Solar Meteorological Observation Specification

Field name	Unit	Source	Required horizons	Definition / forecast use
poa_wm2	W/m2	POA sensor	5min to 2hour	Plane-of-array irradiance; primary solar power driver.
ghi_wm2	W/m2	GHI sensor	15min to day-ahead	Global horizontal irradiance; used for conversion and validation.
ambient_temp_c	deg C	met station	2hour to day-ahead	Ambient temperature for PV efficiency and NWP comparison.
module_temp_c	deg C	module sensor	5min to 2hour	Module temperature; high values reduce PV output.
cloud_cover_pct	%	met/NWP	15min to day-ahead	Cloud cover driver for ramp and day-ahead uncertainty.

Field name	Unit	Source	Required horizons	Definition / forecast use
cloud_obstruction_probabil ity	0-1	sky-camera proxy	5min,15min	Early warning signal for cloud ramp-up/down.
sky_camera_cloud_motion	vector	sky imager	5min,15min	Cloud motion vector; gives lead time before output change.

5. NWP Ensemble Specification

Field name	Unit	Refresh	Required horizons	Definition / forecast use
nwp_ghi_forecast_wm2	W/m2	NWP issue cycle	2hour,4hour,40hour,8day	Weather forecast irradiance input.
nwp_cloud_cover_pct	%	NWP issue cycle	2hour,4hour,40hour,8day	Cloud uncertainty and ramp-risk driver.
nwp_temperature_c	deg C	NWP issue cycle	2hour,40hour,8day	Temperature input for PV model and calibration.
nwp_ensemble_member_id	ID	NWP issue cycle	40hour,8day	Needed for probabilistic ensemble quantiles and rank histogram.
nwp_issue_time	timesta mp	NWP issue cycle	40hour,8day	Forecast run time; required to evaluate forecast lead time.
nwp_valid_time	timesta mp	NWP issue cycle	all NWP horizons	Forecast target time; required to align with actuals.

6. Market/Grid and Business Fields

Field	Unit	Required for	Why needed
demand_forecast_mw	MW	dynamic reserve and grid dispatch	Forecast demand is needed when renewable forecast feeds balancing and reserve decisions.
reserve_requirement_mw	MW	dynamic reserve calculation	Allows forecast uncertainty to be translated into reserve requirement.
deviation_penalty_rate	currency/MWh or rule	deviation penalty reduction	Needed to convert forecast error into business cost.
grid_constraint_flag	boolean/enum	curtailment diagnosis	Marks known grid limitation periods.
dispatch_block_id	ID	schedule/deviation	Aligns schedule and actuals to market/dispatch intervals.

7. Forecast Output Schema

Output field	Unit	Description
forecast_issue_time	timestamp	When the forecast is generated.
valid_time	timestamp	Timestamp for which forecast applies.
asset_level	enum	grid, plant, hub or block.
asset_id	string	Asset identifier at selected hierarchy level.
horizon	enum	5min, 15min, 30min, 1hour, 2hour, 4hour, day_ahead_40hour, 8day.
forecast_p10_mw	MW	Lower probabilistic bound.
forecast_p50_mw	MW	Central forecast.
forecast_p90_mw	MW	Upper probabilistic bound.
event_probability	0-1	Probability of ramp/curtailment/data-quality/equipment event where relevant.
method_family	string	Model method used for selected horizon and asset level.
data_quality_flags	array/string	Quality flags used in the forecast.

8. Data Quality Rules

- SCADA freshness: actual_mw and local_limit_mw must refresh within expected interval; stale values should reduce confidence.
- Physical consistency: possible_power_mw should normally be greater than or equal to actual_mw unless there is a model/sensor issue.
- Curtailment rule: if possible_power_mw > local_limit_mw and actual_mw is near local_limit_mw, classify as curtailment, not weather error.
- Forecast alignment: every NWP forecast must have issue_time and valid_time to support lead-time evaluation.
- Hierarchy rollup: grid/plant/hub values must reconcile with child block sums within tolerance.

9. Sample Row Snippet

```
timestamp,asset_level,asset_id,actual_mw,possible_power_mw,scheduled_mw,local_limit_mw,poa_wm2,module_temp_c,scada_
quality,curtailment_status
2025-01-05T15:20:00,block,PAV_S3_B02,33.4,37.2,34.0,55.0,812.3,46.2,GOOD,NONE
```